

1. Applications of matrices

The solution of simultaneous linear equations is a task frequently occurring in engineering. In electrical engineering the analysis of circuits provides a ready example.

However the simultaneous equations arise, we need to study two things:

- (a) how we can conveniently represent large systems of linear equations
- (b) how we might find the solution of such equations.

We shall discover that knowledge of the theory of matrices is an essential mathematical tool in this area.

Representing simultaneous linear equations

Suppose that we wish to solve the following three equations in three unknowns x_1, x_2 and x_3 :

$$\begin{aligned} 3x_1 + 2x_2 - x_3 &= 3 \\ x_1 - x_2 + x_3 &= 4 \\ 2x_1 + 3x_2 + 4x_3 &= 5 \end{aligned}$$

We can isolate three facets of this system: the **coefficients** of x_1, x_2, x_3 ; the **unknowns** x_1, x_2, x_3 ; and the **numbers** on the right-hand sides.

Notice that in the system

$$\begin{aligned} 3x + 2y - z &= 3 \\ x - y + z &= 4 \\ 2x + 3y + 4z &= 5 \end{aligned}$$

the only difference from the first system is the names given to the unknowns. It can be checked that the first system has the solution $x_1 = 2, x_2 = -1, x_3 = 1$. The second system therefore has the solution $x = 2, y = -1, z = 1$.

We can isolate the three facets of the first system by using **arrays** of numbers and of unknowns:

$$\begin{bmatrix} 3 & 2 & -1 \\ 1 & -1 & 1 \\ 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$$

Even more conveniently we represent the arrays with letters (usually capital letters)

$$AX = B$$

Here, to be explicit, we write

$$A = \begin{bmatrix} 3 & 2 & -1 \\ 1 & -1 & 1 \\ 2 & 3 & 4 \end{bmatrix} \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad B = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$$

Here A is called the **matrix of coefficients**, X is called the **matrix of unknowns** and B is called the **matrix of constants**.

If we now append to A the column of right-hand sides we obtain the **augmented matrix** for the system:

$$\left[\begin{array}{ccc|c} 3 & 2 & -1 & 3 \\ 1 & -1 & 1 & 4 \\ 2 & 3 & 4 & 5 \end{array} \right]$$

The order of the entries, or elements, is crucial. For example, all the entries in the second row relate to the second equation, the entries in column 1 are the coefficients of the unknown x_1 , and those in the last column are the constants on the right-hand sides of the equations.

In particular, the entry in row 2 column 3 is the coefficient of x_3 in equation 2.

Representing networks

Shortest-distance problems are important in communications study. Figure 1 illustrates schematically a system of four towns connected by a set of roads.

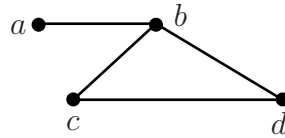


Figure 1

The system can be represented by the matrix

$$\begin{array}{c} a \quad b \quad c \quad d \\ \begin{array}{c} a \\ b \\ c \\ d \end{array} \left[\begin{array}{cccc} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{array} \right] \end{array}$$

The row refers to the town from which the road starts and the column refers to the town where the road ends. An entry of 1 indicates that two towns are directly connected by a road (for example b and d) and an entry of zero indicates that there is no direct road (for example a and c). Of course, if there is a road from b to d (say) it is also a road from d to b .

In this Section we shall develop some basic ideas about matrices.

2. Definitions

An array of numbers, rectangular in shape, is called a **matrix**. The first matrix below has 3 rows and 2 columns and is said to be a '3 by 2' matrix (written 3×2). The second matrix is a '2 by 4' matrix (written 2×4).

$$\left[\begin{array}{cc} 1 & 4 \\ -2 & 3 \\ 2 & 1 \end{array} \right] \quad \left[\begin{array}{cccc} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 9 \end{array} \right]$$

The general 3×3 matrix can be written

$$A = \left[\begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{array} \right]$$

1.4 applications to curve fitting and circuits

We explore a few fun simple examples in this section. I don't intend for you to master the outs and in's of circuit analysis, those examples are for site-seeing purposes.².

Example 1.4.1. Find a polynomial $P(x)$ whose graph $y = P(x)$ fits through the points $(0, -2.7)$, $(2, -4.5)$ and $(1, 0.97)$. We expect a quadratic polynomial will do nicely here: let A, B, C be the coefficients so $P(x) = Ax^2 + Bx + C$. Plug in the data,

$$\begin{array}{rcl} P(0) & = & C = -2.7 \\ P(2) & = & 4A + 2B + C = -4.5 \\ P(1) & = & A + B + C = 0.97 \end{array} \quad \Rightarrow \quad \left[\begin{array}{ccc|c} A & B & C & \\ \hline 0 & 0 & 1 & -2.7 \\ 4 & 2 & 1 & -4.5 \\ 1 & 1 & 1 & 0.97 \end{array} \right]$$

I put in the A, B, C labels just to emphasize the form of the augmented matrix. We can then perform Gaussian elimination on the matrix (I omit the details) to solve the system,

$$\text{rref} \left[\begin{array}{ccc|c} 0 & 0 & 1 & -2.7 \\ 4 & 2 & 1 & -4.5 \\ 1 & 1 & 1 & 0.97 \end{array} \right] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & -4.52 \\ 0 & 1 & 0 & 8.14 \\ 0 & 0 & 1 & -2.7 \end{array} \right] \quad \Rightarrow \quad \begin{array}{l} A = -4.52 \\ B = 8.14 \\ C = -2.7 \end{array}$$

The requested polynomial is $\boxed{P(x) = -4.52x^2 + 8.14x - 2.7}$.

Example 1.4.2. Find which cubic polynomial $Q(x)$ have a graph $y = Q(x)$ which fits through the points $(0, -2.7)$, $(2, -4.5)$ and $(1, 0.97)$. Let A, B, C, D be the coefficients of $Q(x) = Ax^3 + Bx^2 + Cx + D$. Plug in the data,

$$\begin{array}{rcl} Q(0) & = & D = -2.7 \\ Q(2) & = & 8A + 4B + 2C + D = -4.5 \\ Q(1) & = & A + B + C + D = 0.97 \end{array} \quad \Rightarrow \quad \left[\begin{array}{cccc|c} A & B & C & D & \\ \hline 0 & 0 & 0 & 1 & -2.7 \\ 8 & 4 & 2 & 1 & -4.5 \\ 1 & 1 & 1 & 1 & 0.97 \end{array} \right]$$

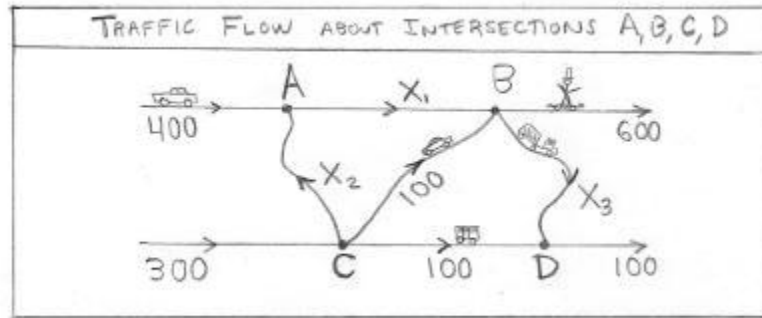
I put in the A, B, C, D labels just to emphasize the form of the augmented matrix. We can then perform Gaussian elimination on the matrix (I omit the details) to solve the system,

$$\text{rref} \left[\begin{array}{cccc|c} 0 & 0 & 0 & 1 & -2.7 \\ 8 & 4 & 2 & 1 & -4.5 \\ 1 & 1 & 1 & 1 & 0.97 \end{array} \right] = \left[\begin{array}{cccc|c} 1 & 0 & -0.5 & 0 & -4.07 \\ 0 & 1 & 1.5 & 0 & 7.69 \\ 0 & 0 & 0 & 1 & -2.7 \end{array} \right] \quad \Rightarrow \quad \begin{array}{l} A = -4.07 + 0.5C \\ B = 7.69 - 1.5C \\ C = C \\ D = -2.7 \end{array}$$

It turns out there is a whole family of cubic polynomials which will do nicely. For each $C \in \mathbb{R}$ the polynomial is $\boxed{Q_C(x) = (c - 4.07)x^3 + (7.69 - 1.5C)x^2 + Cx - 2.7}$ fits the given points. We asked a question and found that it had infinitely many answers. Notice the choice $C = 4.07$ gets us back to the last example, in that case $Q_C(x)$ is not really a cubic polynomial.

²...well, modulo that homework I asked you to do, but it's not that hard, even a Sagittarian could do it.

Example 1.4.3. Consider the following traffic-flow pattern. The diagram indicates the flow of cars between the intersections A, B, C, D. Our goal is to analyze the flow and determine the missing pieces of the puzzle, what are the flow-rates x_1, x_2, x_3 . We assume all the given numbers are cars per hour, but we omit the units to reduce clutter in the equations.



We model this by one simple principle: conservation of vehicles

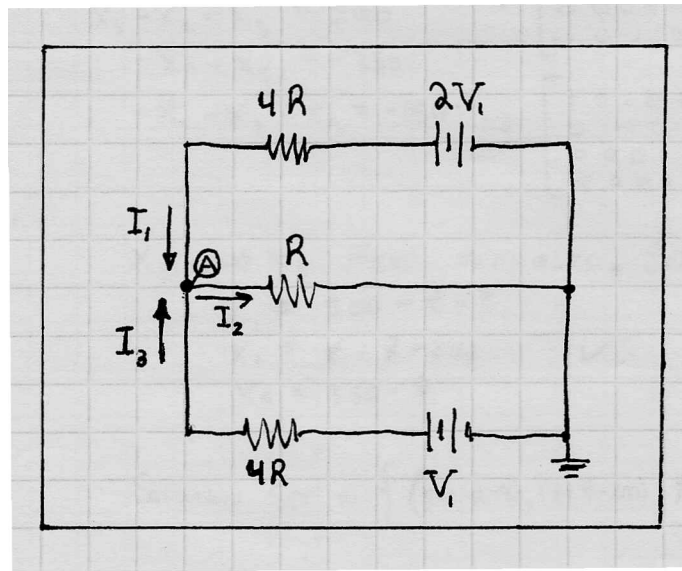
$$\begin{aligned} A: \quad x_1 - x_2 - 400 &= 0 \\ B: \quad -x_1 + 600 - 100 + x_3 &= 0 \\ C: \quad -300 + 100 + 100 + x_2 &= 0 \\ D: \quad -100 + 100 + x_3 &= 0 \end{aligned}$$

This gives us the augmented-coefficient matrix and Gaussian elimination that follows (we have to rearrange the equations to put the constants on the right and the variables on the left before we translate to matrix form)

$$\text{rref} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 400 \\ -1 & 0 & 1 & -500 \\ 0 & 1 & 0 & 100 \\ 0 & 0 & 1 & 0 \end{array} \right] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 500 \\ 0 & 1 & 0 & 100 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

From this we conclude, $x_3 = 0, x_2 = 100, x_1 = 500$. By the way, this sort of system is called **overdetermined** because we have more equations than unknowns. If such a system is consistent they're often easy to solve. In truth, the rref business is completely unnecessary here. I'm just trying to illustrate what can happen.

Example 1.4.4. Let $R = 1\Omega$ and $V_1 = 8V$. Determine the voltage V_A and currents I_1, I_2, I_3 flowing in the circuit as pictured below:



Conservation of charge implies the sum of currents into a node must equal the sum of the currents flowing out of the node. We use Ohm's Law $V = IR$ to set-up the currents, here V should be the voltage dropped across the resistor R .

$$I_1 = \frac{2V_1 - V_A}{4R} \quad \text{Ohm's Law}$$

$$I_2 = \frac{V_A}{R} \quad \text{Ohm's Law}$$

$$I_3 = \frac{V_1 - V_A}{4R} \quad \text{Ohm's Law}$$

$$I_2 = I_1 + I_3 \quad \text{Conservation of Charge at node A}$$

Substitute the first three equations into the fourth to obtain

$$\frac{V_A}{R} = \frac{2V_1 - V_A}{4R} + \frac{V_1 - V_A}{4R}$$

Multiply by $4R$ and we find

$$4V_A = 2V_1 - V_A + V_1 - V_A \Rightarrow 6V_A = 3V_1 \Rightarrow V_A = V_1/2 = 4V.$$

Substituting back into the Ohm's Law equations we determine $I_1 = \frac{16V - 4V}{4\Omega} = 3A$, $I_2 = \frac{4V}{1\Omega} = 4A$ and $I_3 = \frac{8V - 4V}{4\Omega} = 1A$. This obviously checks with $I_2 = I_1 + I_3$. In practice it's not always best to use the full-power of the rref.